Apply deductive reasoning to proofs involving shapes in the plane and use theorems to solve spatial problems

Learning intention

- apply deductive reasoning to proofs involving shapes in the plane and use theorems to solve spatial problems.

Teach and model

- distinguishing between a practical demonstration and a proof.
- investigating proofs of geometric theorems and using them to solve spatial problems.
- applying an understanding of relationships to deduce properties of geometric figures.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.